

Understanding dSR-RL: Search Result Disambiguation



Problem Overview

- Search engines fail with ambiguous queries (e.g., 'Jackson').
- Users manually sift through unrelated results.
- Arabic presents added complexity: diacritics, orthographic variation.





Why Static Clustering Fails

- Fixed parameters; no adaptation to user intent.
- No feedback loop to improve result organization.
- Lack of Arabic language support in many models.

Our Solution – DSR-RL

- Sentence-BERT + BERTopic + RL.
- Clusters search results based on semantic meaning.
- Learns from user feedback: merge, split, rate.



System Workflow

- 1. Preprocess query/snippets.
- 2. Generate embeddings (BERT or TF-IDF).
- 3. Apply clustering algorithm.
- 4. Show clustered results in UI.
- 5. Gather feedback and adjust via RL agent.



Text Preprocessing

- Arabic normalization: remove diacritics, unify Hamza/Alif.
- Tokenization via Farasa/spaCy.
- Light stemming reduces Arabic vocab size by 23%.



Clustering Techniques

- Five options: K-Means, DBSCAN, HDBSCAN, GMM, BERTopic.
- BERTopic preferred for topic labeling and interpretability.
- RL agent switches based on data density/structure.



Default Clustering Parameters

Algorithm	Key Param	Value	Library
K-Means	k	elbow	scikit-learn
DBSCAN	ϵ	0.7	scikit-learn
HDBSCAN	min samples	5	hdbscan
GMM	components	k	scikit-learn
BERTopic	min size	10	BERTopic



Reinforcement Learning Setup

- State: query length, vector density, sparsity, divergence.
- Action: choose (embedding, clustering) combo.
- Reward: cluster quality + user feedback actions (merge/split).

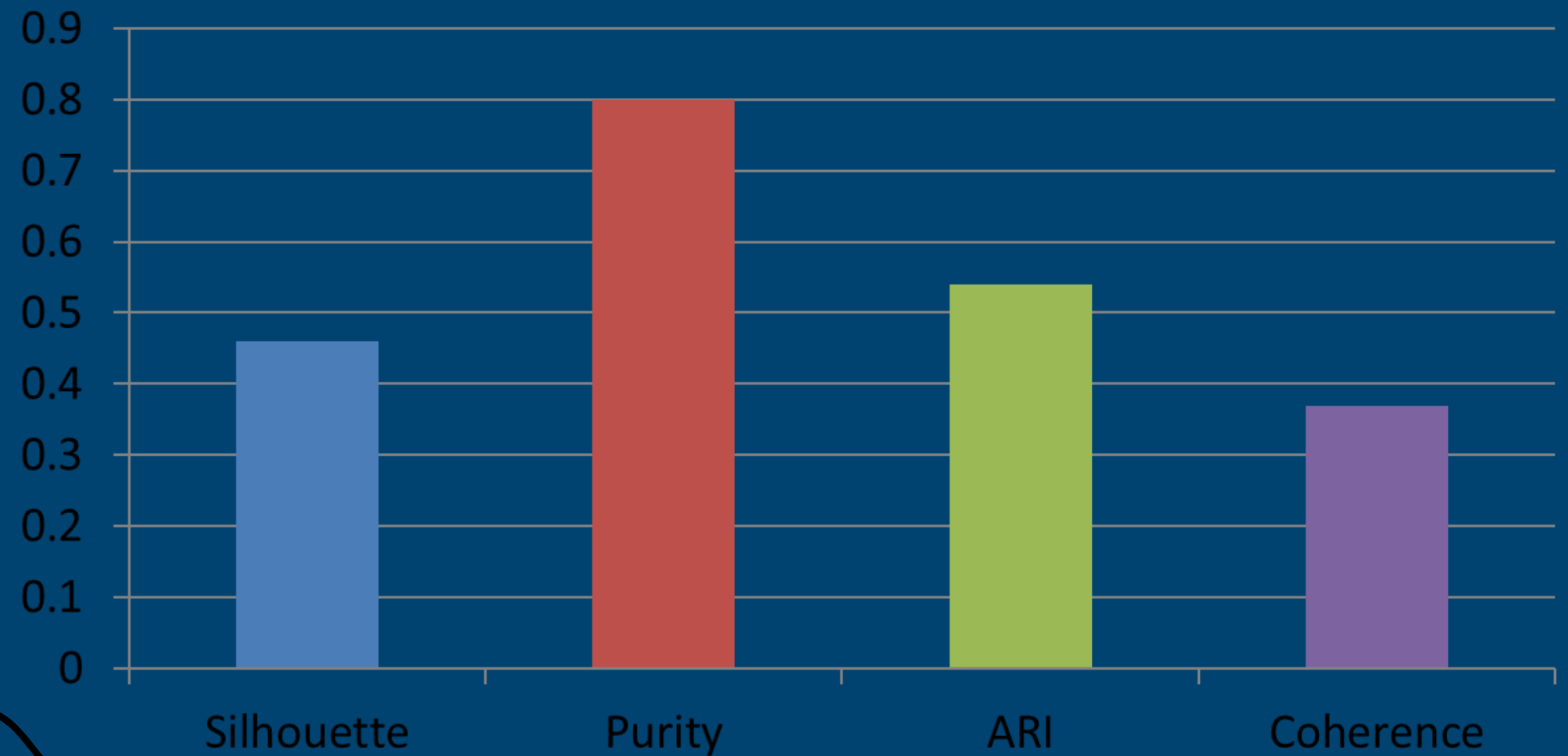


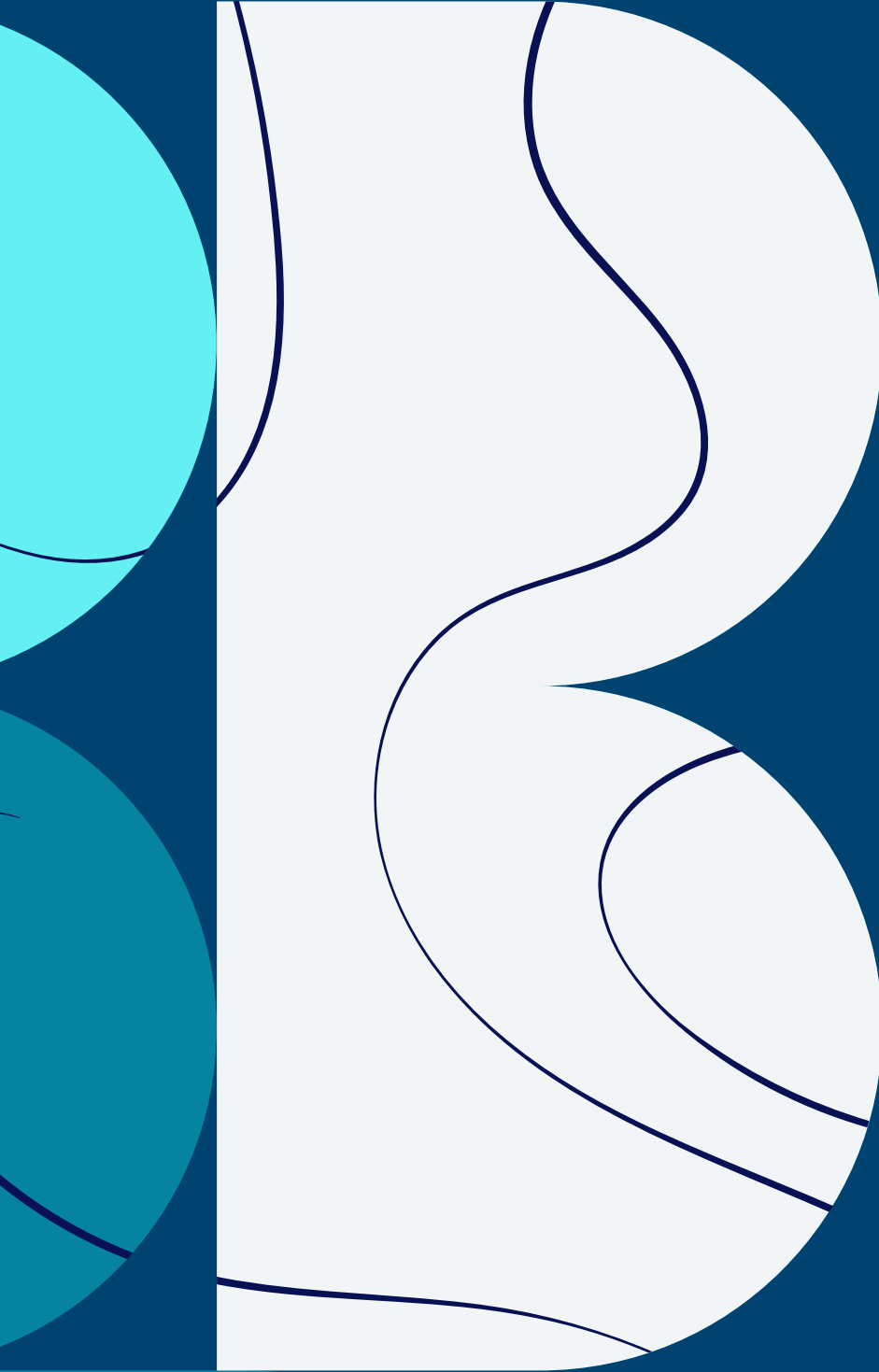
Clustering Evaluation Metrics

Metric	Static BERTopic	DSR-RL
Silhouette	0.38	0.46
Purity	0.71	0.80
Adjusted Rand Index (ARI)	0.44	0.54
Topic Coherence	0.29	0.37

Metric Comparison

DSR-RL vs Static





Experimental Setup

- Datasets: TREC Web Diversity, MIRACL-Arabia, Wikipedia.
 - CPU: Intel i7, 16GB RAM, no GPU.
 - Embedding: 480ms, Clustering: 220ms, Total latency: ~920ms.
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Raw List of Results (Before Clustering)

The screenshot shows a web browser window with two tabs: 'Dynamic Search Result Disamb' and 'localhost:5000'. The page is divided into two main sections: 'Original Search Results' and 'AI-Clustered Results'.

Original Search Results:

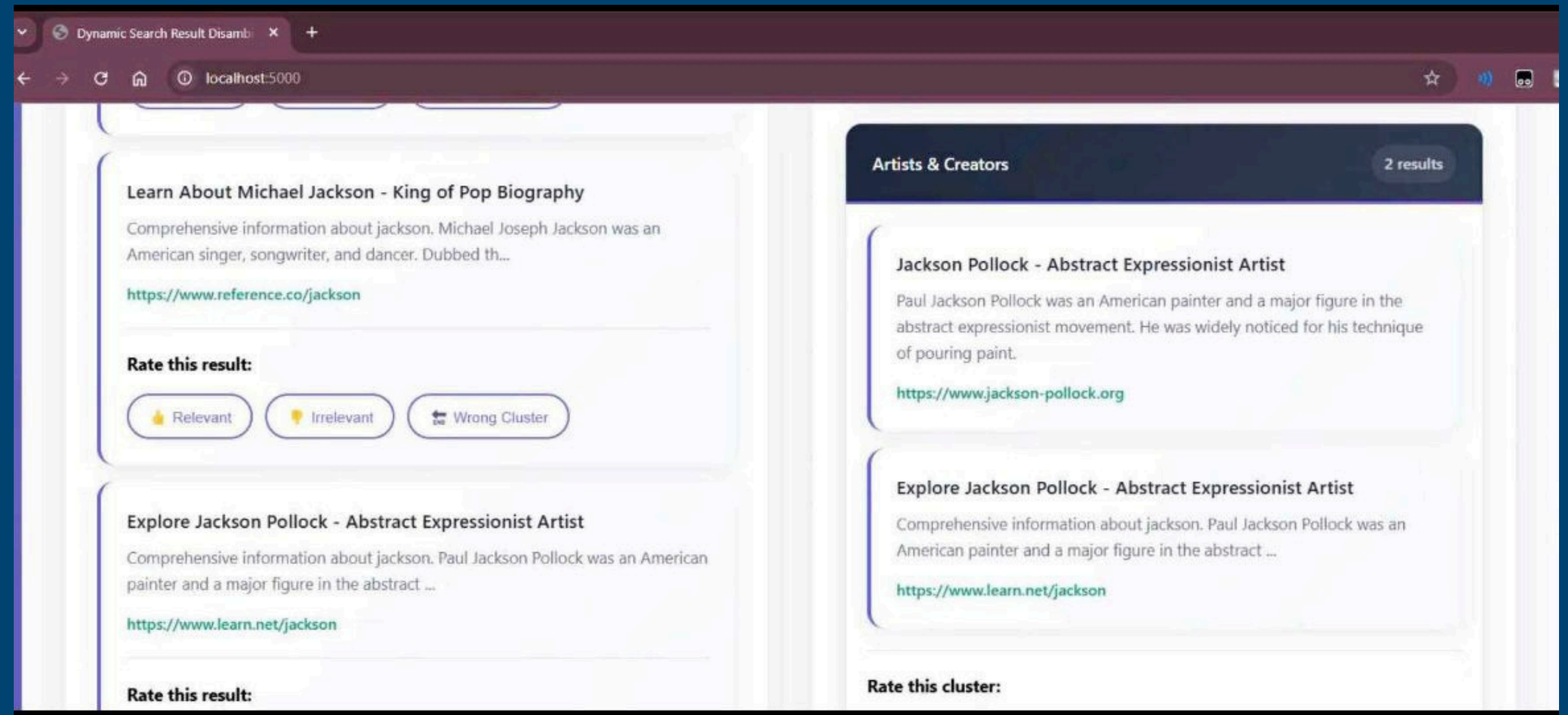
- Michael Jackson - King of Pop Biography**
Michael Joseph Jackson was an American singer, songwriter, and dancer. Dubbed the "King of Pop", he is regarded as one of the most significant cultural figures of the 20th century.
<https://www.michaeljackson.com/biography>
- Rate this result:**
Buttons: Relevant, Irrelevant, Wrong Cluster
- Jackson, Mississippi - Capital City Guide**
Jackson is the capital and most populous city of Mississippi. The city serves as the anchor for the Jackson metropolitan statistical area.
<https://www.jacksonms.gov>

AI-Clustered Results:

- General Information** (12 results)
- Jackson, Mississippi - Capital City Guide**
Jackson is the capital and most populous city of Mississippi. The city serves as the anchor for the Jackson metropolitan statistical area.
<https://www.jacksonms.gov>
- Andrew Jackson - 7th President of the United States**
Andrew Jackson was the seventh President of the United States from 1829 to 1837. Before being elected president, Jackson gained fame as a general.
<https://www.whitehouse.gov/about-the-white-house/presidents/andrew-jackson>

Jackson_raw list

Clustered Search Results



Jackson_clustered

Conclusion & Future Work

- DSR-RL improves clustering without labeled data.
- Handles Arabic and ambiguous queries dynamically.
- Future work: integrate image/audio evidence, optimize for mobile.

Thank You



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[Click here to show the demo
video and the codes](#)